01 / PROBLEM AND CONTRIBUTION

IoT energy-billing demonstrator

IoT mini-project · Manipal Institute of Technology | November 2023 | Two-person team

RESULT AT A GLANCE

Demonstrated cloud-connected net-energy billing calculations from example consumption and generation inputs using an ESP8266 and a fixed tariff.

The engineering question

How can energy-use and generation values be processed on a microcontroller and displayed remotely?

Situation

Energy consumption and generation values can be combined into a net-value calculation and displayed remotely. This demonstrator uses substituted firmware inputs to exercise that calculation and cloud connection.

Task

Work with the team to implement the calculation on an ESP8266 and display the result through Arduino IoT Cloud.

My contribution

Completed collaboratively with B. K. V. Surya, combining the firmware calculation with an Arduino IoT Cloud demonstration.

Tools and concepts

ESP8266 / Arduino IoT Cloud / Embedded C / IoT / Data visualization

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

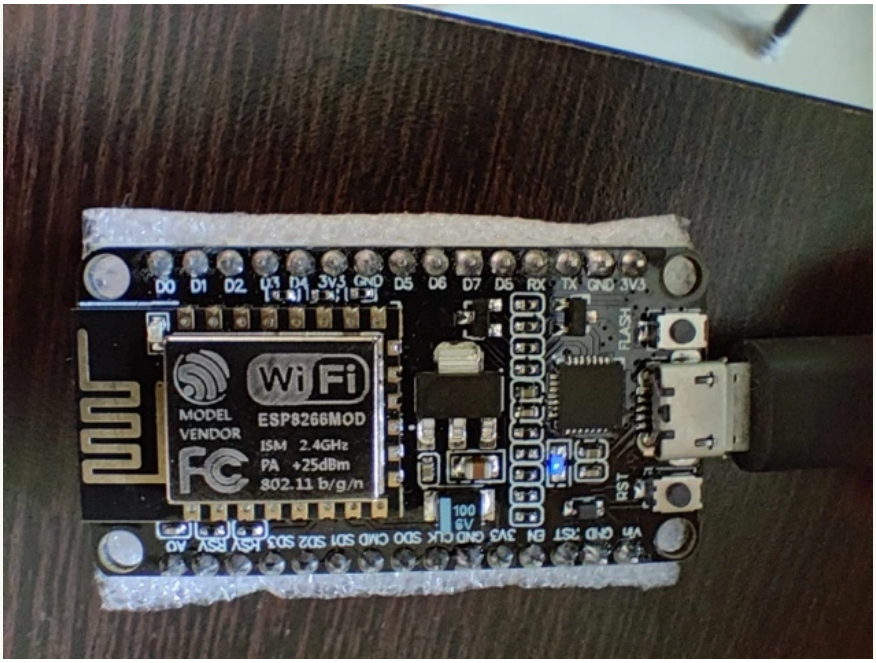
Used example consumption and generation values in the firmware, calculated their difference, and applied a fixed price-per-unit variable.

02 / Implementation step

Configured the ESP8266 connection to Arduino IoT Cloud and updated the displayed result from the main loop. The report shows different sample consumption/generation cases on web and mobile views.

Calculation and presentation

The firmware subtracts the example generation input from the example consumption input and applies a fixed price-per-unit value. Arduino IoT Cloud presents the resulting demonstration values remotely.



ESP8266 hardware

ESP8266 board photographed in the IoT project report; the project uses sample input values.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Reported result

The implemented code demonstrates a cloud-connected calculation using substituted input values. Physical measurement modules are listed as a proposed improvement.

Demonstration versus measurement

The supplied code substitutes input values rather than acquiring verified energy measurements. Physical metering, units, integration intervals, and export handling must be established before the demonstration can be described as a billing instrument.

Scope and limitations

The demonstrated firmware uses sample inputs and a fixed tariff. Connecting physical energy sensors and distinguishing consumption from export are the next steps toward a metering application.

Possible next steps - proposed, not completed results

- Integrate physical energy measurement and document units and sampling/integration intervals.
- Separate import and export cases and validate tariff handling against defined examples.

Source material: IOT_project[1].docx, implementation code, explanation, and scope of improvement.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: IoT-Based-Smart-grid-along-with-auto-billing-